

Harshwardhan Fartale

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Work Experience

ZS

Advanced Data Science Associate

October 2025 - Present

- Built an agentic AI-powered regulatory platform for life sciences that automated pre-submission FDA dossier review (filing compliance scoring, correspondence verification, guidance gap analysis, and competitor benchmarking) with word-level traceability across 2000+ source documents, cutting submission-readiness effort from months of manual cross-referencing to a single automated run.

DRDO & (IISC) Indian Institute of Science, Bangalore

Predoc Associate

July 2024 - September 2025

- Wrote C++ inference modules for CASDIC combat-aircraft hardware, optimizing ML model execution under embedded latency constraints and reducing inference time by 35% versus baseline.
- Fine-tuned IBM Granite and Gemma models for scientific ML tasks, built a Graph-RAG pipeline that outperformed baselines by 45%, and led development of PULSE and SciREX under AiREX Lab.

Bajaj Finserv Health

Data Scientist

Jan 2024 - June 2024

- Built a fraud detection model that caught 20% more abusive transactions than the existing rule-based system, processing 3000+ transactions/day across Bajaj Health's payments pipeline.
- Built a graph-based fraud detection pipeline in Neo4j and Memgraph with optimized SQL feature extraction, improving linked-fraud recall by 55% and model precision by 30% versus tabular baselines.

OpenMined

Open Source Technical Writer

Jan 2023 - June 2023

- Wrote beginner tutorials and technical documentation on differential privacy and federated learning for PySyft, and integrated them into OpenMined's onboarding docs.
- Published blog posts explaining privacy-preserving ML concepts, driving monthly reads across OpenMined channels.

Publications

Can Linguistically Related Languages Guide LLM Translation in Low-Resource Settings?

LoResMT: EACL Workshop (2026)

The Evolution of FlashAttention

ICLR Blog Post Track (peer-reviewed) (2026)

Disentangling Recall and Reasoning in Transformer Models through Layer-wise Attention and Activation Analysis

AAAI XAI4Science Workshop (2026)

RADAR: Mechanistic Pathways for Detecting Data Contamination in LLM Evaluation

NeurIPS LLM Evaluation Workshop (2025)

Equilibrium Dynamics and Mitigation of Gender Bias in Synthetically Generated Data Evaluation

AAAI Workshop on Shaping Responsible Synthetic Data in the Era of Foundation Models (2025)

Skills

Languages

Python, C++, SQL, Triton

ML/AI

PyTorch, Scikit-learn, RAG pipelines, LLM fine-tuning, Graph-RAG

Infrastructure

Docker, Kubernetes, GCP, FastAPI, Neo4j, Memgraph, PostgreSQL

Technical Writing

Medium, Hashnode, Substack

Key Projects

Alcuin — Deep AI Learning Platform

Founder & Engineer

2025 – Present

- Built a learning platform with mathematically rigorous AI courses and structured topic flows for many simultaneous learners.

Shroud — Identity Infrastructure for AI Agents

Founder & Engineer

2025 – Present

- Built identity infrastructure for AI agents using disposable emails, phone numbers, and policy-constrained cards for verification.

Education

National Institute of Technology, Hamirpur

CGPI – 8.56

B.Tech in Electrical Engineering, GATE Data Science (DA) AIR 663

Dec 2020 – May 2024

Additional Experience

IISc Bangalore

Teaching Assistant, DS-246 Generative & Agentic AI in Practice

Jul 2025 – Sep 2025

DRDO Industry Academia Centres of Excellence (DIA-CoEs)

ML & Data Science Trainer for DRDO Scientists

Jan 2025